Executive Summary: COVID-19 Global Impact & Vaccination Analysis

Prepared By: Mohammad Samreen

Organization: Veda Technologies

Data Source: Our World in Data (Compact COVID-19 Dataset)

Project Objective

This analysis evaluates the global trajectory of the COVID-19 pandemic, focusing specifically on the correlation between vaccination rollout speeds and the subsequent impact on case fatality rates (CFR) and total mortality.

Methodology & Data Engineering

The underlying dataset was processed using a custom Python pipeline to ensure data integrity and standardize reporting metrics. Key operations included:

- **Data Cleansing:** Filtered out aggregate regional data to isolate country-level metrics and resolved missing daily reporting values using forward-filling and zero-assumption techniques.
- **Feature Engineering:** Derived 7-day rolling averages to smooth weekend reporting anomalies, calculated the Case Fatality Rate (CFR%), and engineered population-normalized metrics (per million) to allow for equitable benchmarking between countries of vastly different sizes.
- **Visualization:** Deployed an interactive Power BI dashboard featuring dynamic date and country parameters, wave-tracking line charts, and dual-axis comparative columns.

Key Analytical Insights

1. **The Mortality Decoupling Effect:** The most prominent finding across the data is the stark decoupling of case volumes from mortality rates following mass immunization. While new case waves continued to spike in late 2021 and 2022, the secondary Y-axis tracking the 7-day average of deaths remained significantly suppressed once a country's fully vaccinated population surpassed the 50–60% threshold.
2. **CFR Stabilization:** Early pandemic phases (2020) show highly volatile and elevated Case Fatality Rates due to a completely naive population and lack of treatments. As vaccination bars consistently rise on the dual-axis chart, the CFR line stabilizes at a structurally lower baseline.
3. **Equitable Benchmarking via Normalization:** Relying strictly on raw death totals misrepresents the pandemic's impact. Utilizing the engineered "Total Deaths Per Million" metric reveals that several smaller nations experienced far more severe impacts relative to their population size than the raw leaders in absolute fatalities.